

Global RF for Wildfire Susceptibility in Liguria

Pierre Hauptmann
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1 Introduction

Liguria, a narrow coastal region in northwestern Italy, is prone to wildfires due to steep slopes, mediterranean vegetation, and urban-forest interfaces. Hotter summers and prolonged droughts have increased fire risk. This study applies a Random Forest classifier to predict wildfire occurrence. We use 10 000 presence/absence points, rasters predictors (elevation, distances, aspect, climate, vegetation), and burned-area shapefiles. Our goal is to produce a probability map, evaluate model accuracy, and identify key factors influencing fire risk.

2 Method

2.1 Data Sources

- **Points:** WF_pa_10K.csv – 10 000 sample points with WF = 1 (fire) or 0 (no fire), loaded as `sf` and reprojected to EPSG:32632.
- **Shapefiles:** BurnedArea.shp (historical perimeters), Liguria.shp (boundary).
- **Rasters** (10 m resolution):
 - dem.tif (elevation), slope.tif, eastness.tif, northness.tif.
 - dist_roads.tif, dist_pathways.tif, dist_urban.tif.
 - maenTemperature.tif (mean annual temperature), meanPrecipitation.tif (mean annual precipitation), Vegetation.tif (vegetation code).

2.2 Exploratory Analysis

We extracted raster values at all points and computed a Pearson correlation matrix. Distance to roads and pathways had high correlation ($r \approx 0.85$); eastness/northness were nearly uncorrelated with other predictors. A heatmap ensured no extreme multicollinearity.

2.3 Spatial Blocking & Partitioning

Using `blockCV`, we created five spatial blocks (10 km each). The largest block (2307 points) served as the

test set; the remaining 7698 points were split into a training subset (80%, 6158 pts) and validation subset (20%, 1540 pts). This preserves spatial independence between train/validation and test.

2.4 Data Preparation

A helper function `prep_df(sf_df)` extracts raster values via `terra::extract()` and returns a data frame with `presence` (factor 0/1) plus predictors. We created `train_df`, `valid_df`, and `test_df`.

2.5 Random Forest Model

We used the `mlr3` framework with the `ranger` learner (`classif.ranger`) to construct our Random Forest classifier. The model was configured with 500 trees and impurity-based variable importance, and it was set to return class-“1” probabilities by specifying `predict_type="prob"`. First, we performed a 5-fold cross-validation on the training dataset to assess model performance in terms of accuracy, AUC, and Brier score. After confirming satisfactory cross-validation results, we retrained the final Random Forest on the combined training and validation data (`train_df` + `valid_df`, totaling 7698 points). Finally, we applied this trained model to the held-out test dataset (`test_df`, consisting of 2307 points) and calculated the confusion matrix, test-set AUC, and other relevant metrics to evaluate how well the model generalized to new data.

2.6 Metrics & Outputs

- **Metrics:** Confusion matrix, accuracy, sensitivity, specificity, AUC.
- **Variable Importance:** Top 10 predictors.
- **Partial Dependence Plots:** For vegetation code, elevation, and precipitation.
- **Probability Map:** Applied final model to entire raster stack, producing `wildfire_susceptibility.tif`.

3 Results

3.1 Cross-Validation

Metric	5-fold CV
Accuracy	0.7805
AUC	0.8624
Brier Score	0.1539

Table 1: 5-fold CV performance on training set.

CV yielded accuracy 78%, AUC = 0.862, and Brier score 0.154, indicating good discrimination and calibration (Liaw & Wiener, 2002).

3.2 Test Set Evaluation

Actual	Predicted		Total
	No Fire (0)	Fire (1)	
No Fire (0)	963	337	1300
Fire (1)	254	753	1007
Total	1217	1090	2307

Table 2: Confusion matrix on test set (2307 points).

On the test set, the model achieved an overall accuracy of 0.75, with a sensitivity (recall for fires) of 0.75 and a specificity of 0.74. The test-set AUC was 0.8075, indicating that the model generalizes well to unseen data.

3.3 ROC Curves

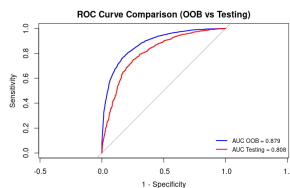


Figure 1: ROC: OOB (blue) vs. Test (red).

OOB AUC = 0.88, Test AUC = 0.8075. The small gap indicates minimal overfitting (Abatzoglou & Williams, 2016).

3.4 Variable Importance

Rank	Variable	Importance
1	veg_code	610
2	meanPrecipitation	327
3	maenTemperature	305
4	dem (elevation)	180
5	dist_roads	95
6	dist_pathways	90
7	slope	55
8	dist_urban	30
9	eastness	12
10	northness	8

Table 3: Top 10 predictor importance (impurity).

Vegetation type is the dominant predictor; codes 40–75 (Mediterranean shrubs/pine) have highest risk. Precipitation and temperature follow, since low rainfall and high heat dry fuels (Bradstock, 2010). Elevation ranks fourth, reflecting climate and human access influences (Krawchuk & Moritz, 2011).

3.5 Partial Dependence Plots

- **Vegetation Code:** Fire probability peaks 0.7 for codes 40–75 (Mediterranean shrub/pine), versus <0.3 for wet deciduous types (Pausas & Keeley, 2009).
- **Elevation (DEM):** Probability rises from sea level to 1000 m (peak 0.57), then declines above 1000 m as cooler, moister conditions prevail [?].
- **Precipitation:** Very low rainfall (50 mm) yields highest risk (0.62); moderate rainfall (100 mm) lowers risk (0.45), while very high rainfall (>120 mm) slightly increases risk due to biomass growth (Krawchuk & Moritz, 2011).

3.6 Final Probability Map

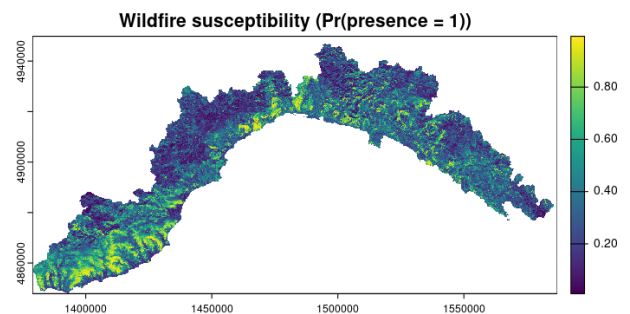


Figure 2: Wildfire susceptibility map (Pr(presence=1)) for Liguria.

- **High risk ($p > 0.7$):** Urban coastal fringes (human ignitions), mid-altitude slopes with flammable shrub/pine, western Liguria (drier climate).
- **Low risk ($p < 0.4$):** Elevations > 1200 m (cooler, wetter), inland mountain valleys with high precipitation.

Patterns align with human ignition near roads/towns and fuel/climate driving fire intensity (Syphard et al., 2007).

The interactive map shows wildfire susceptibility across Liguria. Coastal zones and areas with specific vegetation types exhibit higher risk ($p > 0.7$) because dense vegetation plus proximity to roads and urban areas enhances ignition and spread (Syphard et al., 2007). Inland mountainous regions generally have lower risk ($p < 0.4$) due to cooler temperatures and higher precipitation.

Human activity appears to trigger wildfire ignitions, while environmental factors as vegetation type, temperature, precipitation, and terrain determine fire intensity and spread. Western Liguria's drier climate and vegetation may contribute to its higher susceptibility.

Model improvements could include adding dynamic weather and fuel data. Wind speed and direction would help predict spread rates, and live fuel moisture would indicate ignition likelihood. Lightning strike records and human activity metrics (e.g., campfire permits, visitor numbers) could highlight ignition hotspots. Incorporating seasonal vegetation growth and fine-scale terrain features (e.g., slope aspect) would capture local variations in fire risk.

4 Discussion

4.1 Strengths

The model's test-set AUC of 0.8075 demonstrates reliable identification of high-risk areas. Key predictors (vegetation type, precipitation, and temperature) align with established fire ecology, confirming that flammable fuels combined with dry, hot conditions increase susceptibility (Bradstock, 2010). Spatial blocking preserved geographic independence between training/validation and test sets, reducing overfitting and strengthening evaluation credibility.

4.2 Limitations

To enhance predictive accuracy, the model could incorporate seasonal climate variables, such as monthly precipitation, temperature anomalies, or drought indices—to capture short-term weather patterns that drive fire behavior. Adding wind speed and direction data

would refine spread predictions, and including live fuel moisture measurements would better represent real-time ignition potential. A continuous vegetation index (e.g., NDVI) or a detailed fuel-model classification could provide finer distinctions in fuel flammability. Integrating lightning strike frequency and human activity metrics (e.g., visitor counts or campfire permit records) would help identify ignition hotspots. For future work, exploring time-series approaches such as LSTM networks could model temporal dependencies and improve the detection of seasonal or interannual fire trends.

5 Conclusion

A Random Forest model with environmental predictors and spatial blocking produced a good wildfire susceptibility map for Liguria. Vegetation type, precipitation, and temperature emerged as the strongest factors. Incorporating dynamic weather, fuel moisture, and finer-scale terrain features could further enhance accuracy.

References

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